Lecture 1: Real Numbers

AMA3707 Real Analysis

1 Numbers

Definition 1.1 (Natural Numbers). [def:natural-number] The *natural numbers* are the positive integers:

$$\mathbb{N} = \{1, 2, 3, \ldots\}.$$

Convention 1.1 (Natural Numbers Exclude Zero). [conv:natural-numbers-exclude-zero] $0 \notin \mathbb{N}$.

Definition 1.2 (Integers). [def:integer] The *integers* are the whole numbers, including the positive integers, the negative integers, and zero:

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

Definition 1.3 (Rational Numbers). [def:rational-numbers] A real number is called *rational* if it can be expressed as a quotient of two integers. The set of rational numbers is

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\}.$$

Proposition 1.1 (Unique Reduced Form). [prop:unique-reduced-rational-form] *For every $x \in \mathbb{Q}$, there exists a unique pair $p, q \in \mathbb{Z}$ such that*

$$x = \frac{p}{q}, \qquad q > 0, \qquad \gcd(|p|, q) = 1.$$

Remark 1.1. [rem:number-set-inclusions] Under the usual identifications, $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q}$.

2 Fields

Definition 2.1 (Closed Operation). [def:closed-operation] An operation $*$ is said to be *closed* on a set A if for all $a, b \in A$, we have $a * b \in A$.

Definition 2.2 (Field). [def:field] A *field* is a set F equipped with addition and multiplication such that both operations are multiplication are

- closed operations on F,
- associative, meaning for all $a, b, c \in F$, we have $(a+b)+c = a+(b+c)$ and $(a\cdot b)\cdot c = a\cdot(b\cdot c)$,
- commutative, meaning for all $a, b \in F$, we have $a + b = b + a$ and $a \cdot b = b \cdot a$,
- have identity elements 0 and 1,
- each element has an additive inverse, such that for each $a \in F$, there exists $-a \in F$ with $a + (-a) = 0$,
- each non-zero element has a multiplicative inverse, such that for each $a \in F$ with $a \neq 0$, there exists $a^{-1} \in F$ with $a \cdot a^{-1} = 1$

Proposition 2.1 ($\mathbb{Q}$ Is a Field). [prop:rationals-are-a-field] *The rational numbers with their usual addition and multiplication form a field.*

Remark 2.1 ($\mathbb{N}$ and $\mathbb{Z}$ Are Not Fields). [rem:natural-and-integers-not-fields] The natural numbers and the integers are not fields because they do not have multiplicative inverses for every non-zero element.

For a counter example, consider the integer 2. There is no integer x such that $2 \cdot x = 1$

3 Ordered Set

Definition 3.1 (Ordered Set). [def:ordered-set] A set F is *ordered* if it is transitive and reflexive and

$$\forall x, y \in F, \text{ either one of the following happens } x \leq y, x = y, \text{ or } y \leq x$$

It is also called partial ordered.

Definition 3.2 (Transitive). [def:transitive] An ordering on a set is *transitive* if

$$\forall x, y, z \in F, \text{ if } x \leq y \text{ and } y \leq z, \text{ then } x \leq z$$

Definition 3.3 (Reflexive). [def:reflexive] An ordering on a set is *reflexive* if

$$\forall x \in F, x = x$$

4 Irrational Numbers

Lemma 4.1 (Parity of a Square). [lem:square-parity] *For every $n \in \mathbb{Z}$,*

$$n^2 \text{ is even} \iff n \text{ is even},$$

and equivalently,

$$n^2 \text{ is odd} \iff n \text{ is odd}.$$

Proposition 4.1 (No Rational Square Root of 2). [prop:no-rational-square-root-of-two] *There is no $q \in \mathbb{Q}$ such that $q^2 = 2$.*

Proof. Suppose, for contradiction, that some $q \in \mathbb{Q}$ satisfies $q^2 = 2$.
According to the proposition 1.1, and $q \in \mathbb{Q}$

$$\exists p, r \in \mathbb{Z}, q = \frac{p}{r}, r > 0, \gcd(|p|, r) = 1$$

Then we have

$$\left(\frac{p}{r}\right)^2 = 2 \implies p^2 = 2r^2$$

Since p^2 is even, the square-parity lemma implies that p is even. Therefore,

$$\exists k \in \mathbb{Z}, \quad p = 2k.$$

So,

$$(2k)^2 = 2r^2 \implies 2k^2 = r^2$$

By the same argument, r is also even. Meaning that both p and r are even, which contradicts the assumption that $\gcd(|p|, r) = 1$.
So, it contradicts the assumption that $q \in \mathbb{Q}$ satisfies $q^2 = 2$. Therefore, there is no $q \in \mathbb{Q}$ such that $q^2 = 2$. □

Remark 4.1 (Larger Number System). [rem:sqrt-two-larger-number-system] Within $\mathbb{Q}$, the proposition only says that the equation $q^2 = 2$ has no solution $q \in \mathbb{Q}$. Writing $\sqrt{2} \notin \mathbb{Q}$ presupposes a larger number system in which $\sqrt{2}$ exists.

Remark 4.2 (Completeness). [rem:rational-incompleteness] The existence, in a larger number system, of some $x \notin \mathbb{Q}$ satisfying $x^2 = 2$ does not by itself prove that $\mathbb{Q}$ is incomplete. Incompleteness must be demonstrated internally [TODO Pf link to completeness axiom].

Convention 4.1 (Irrational Numbers). [conv:irrational-numbers] We use $\mathbb{I}$ to denote the set of *irrational numbers*

Remark 4.3 (Number Sets). [rem:number-sets-inclusions-real] The set of real numbers $\mathbb{R}$ is the union of the set of rational numbers $\mathbb{Q}$ and the set of irrational numbers $\mathbb{I}$:

$$\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$$

Also, in normal conventions, we have:

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}$$

Proposition 4.2 (Integer–Irrational Root Dichotomy). [prop:nth-root-dichotomy] *For $x, n \in \mathbb{N}$ with $n > 1$, $\sqrt[n]{x} \in \mathbb{Z}$ if x is a perfect n-th power, and otherwise $\sqrt[n]{x} \in \mathbb{I}$.*

Remark 4.4 (Sources of Irrational). [rem:sources-of-irrational]
- By proposition 4.2, $\sqrt[n]{x}$ is irrational whenever $x \in \mathbb{N}$, $n > 1$, and x is not a perfect n-th power.
- Taking logarithms
- Constants defined by taking limits, such as π and e.

5 Sets

Definition 5.1 (Intersection). [def:intersection] For sets A and B,

$$x \in A \cap B \iff x \in A \text{ and } x \in B.$$

Definition 5.2 (Union). [def:union] For sets A and B,

$$x \in A \cup B \iff x \in A \text{ or } x \in B.$$

Definition 5.3 (Complement). [def:complement] Given a universal set U and $A \subseteq U$, the complement of A is $A^c = U \setminus A$; equivalently,

$$x \in A^c \iff x \notin A.$$

Definition 5.4 (Set Difference). [def:set-difference] For sets A and B,

$$x \in A \setminus B \iff x \in A \text{ and } x \notin B.$$

Definition 5.5 (Subset). [def:subset] For sets A and B,

$$A \subseteq B \iff \text{for every } x, \quad x \in A \implies x \in B.$$

Definition 5.6 (Set Equality). [def:set-equality] For sets A and B,

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

Theorem 5.1 (De Morgan's Laws). [thm:de-morgan-laws] *For sets A and B in a universal set U,*

$$(A \cap B)^c = A^c \cup B^c \quad \text{and} \quad (A \cup B)^c = A^c \cap B^c.$$

Also,

$$\left(\bigcap_{n=1}^{\infty} A_n \right)^c = \bigcup_{n=1}^{\infty} (A_n)^c \quad \text{and} \quad \left(\bigcup_{n=1}^{\infty} A_n \right)^c = \bigcap_{n=1}^{\infty} (A_n)^c$$

Example 5.1 (Nested Sequence). [ex:nested-sequence] Let $A_n = \{n, n+1, n+2, \dots\}, n \in \mathbb{N}$
ie, $A_1 = \{1, 2, 3, \dots\}, A_5 = \{5, 6, 7, \dots\}$
Trivially we can get $A_1 \subset A_2 \subset \dots$. This is called a nested sequence.
Where

$$\bigcup_{n=1}^{\infty} A_n = A_1 \quad \bigcap_{n=1}^{\infty} = \emptyset$$

The intersection is $\emptyset$ due to if a number, let's say for any natural number p is in the set, then A_{p+1} does not include that number.
So in the universe $\mathbb{N}$,

$$\forall p \notin \bigcap_{n=1}^{\infty} \implies \bigcap_{n=1}^{\infty} = \emptyset$$

6 Mapping

Definition 6.1 (Mapping). [def:mapping] A mapping $f : A \to B$ is a set of rule st.:

$$\forall x \in A, \exists! y = f(x) \in B \leftrightarrow \forall x \in A, \exists! y \in B \text{ st. } (x, y) \in f$$

Definition 6.2 (Domain). [def:domain] The *domain* of a mapping $f : A \to B$ is the set A.

Definition 6.3 (Codomain). [def:codomain] The *codomain* of a mapping $f : A \to B$ is the set B.

Definition 6.4 (Range). [def:range] The *range* of a mapping $f : A \to B$ is a set C such that:

$$C = \{y : \exists x \in A, \text{ st. } y = f(x)\}$$

Remark 6.1 (Range and codomains are different). [rem:range-and-codomains-are-different] The range is a subset of codomain. Notice that some of the values in the codomain is not reachable.
For example,

$$f = x^2 : \mathbb{R} \to \mathbb{R}$$

The codomain is $\mathbb{R}$ but the range is $\mathbb{R}^+ \cup \{0\}$.